QRNG vs Classical PRNG Benchmarking (Revised)

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1. Summary

A single-qubit Hadamard QRNG simulated in Qiskit AerSimulator is compared with Mersenne Twister (MT19937), ChaCha20 and the OS CSPRNG (os.urandom) using five NIST-style tests, entropy estimates and a 256-bit key simulation. **Main result:** at 10,000 and 1,000,000 bits, none of the four sources can be statistically distinguished from the others. Each test fails about 1% of fair streams at alpha = 0.01, which is what the tests are designed to do. MT19937 is statistically fine but cryptographically broken: its full state is recovered from 624 outputs and the next 1,000 outputs are predicted exactly. **Real IBM Quantum hardware results are not included yet** (pending, see section 6).

2. Corrections to the first version of this report

• The ChaCha20 'frequency failure' (statistic 12754) was a **code bug** (statistic computed on bytes, not bits), not an artefact of OS entropy. Fixed; ChaCha20 now uses a real ChaCha20 keystream and passes.
• The 0/1 chi-square equals the monobit statistic squared ($3.24^2 = 10.4976$, same p = 0.0012), so one deviation was counted as two failures. Replaced by a 256-bin byte chi-square.
• AerSimulator draws outcomes from a classical PRNG, so the simulated QRNG is **not true entropy**. Only hardware or a physical source can claim that.
• The earlier QRNG monobit failure (ones-ratio 0.5162, p = 0.0012) was most likely chance on a single 10k-bit stream, not a systematic simulator effect: over 200 fresh 10k-bit streams the simulated QRNG passed monobit 100% of the time.
• For 256-bit keys the expected mean bias is about 0.0249, not 0.0; all sources match this.

3. Method

Sources: (i) QRNG: H gate on |0>, measure, 1000 shots per circuit, in AerSimulator; (ii) MT19937 via Python random; (iii) ChaCha20 keystream (cryptography library, key and nonce from os.urandom); (iv) os.urandom. Tests (fail if p < 0.01): NIST Monobit, Block Frequency (M=128), Runs, DFT Spectral, and a byte-level chi-square (df = 255). Also Shannon entropy (8-bit blocks), min-entropy, autocorrelation, and a 256-bit key simulation (mean bias, LSB balance, uniqueness). Each result is a pass *rate* over many independent streams, because a single run cannot separate a real defect from the expected 1% false-failure rate.

4. Results

4.1 Pass rates, 200 independent streams x 10,000 bits (original sample size)

Test	QRNG (Aer sim)	Mersenne Twister	ChaCha20	os.urandom
Monobit	100%	100%	98%	99%
Block Freq (M=128)	100%	100%	100%	99%
Runs	100%	98%	100%	98%
DFT Spectral	100%	100%	100%	98%
Byte Chi-Square	100%	100%	99%	100%
Mean Shannon (bits/bit)	0.98057	0.98058	0.98060	0.98076
Mean key bias (ideal ~0.0249)	0.02461	0.02528	0.02481	0.02498

4.2 Pass rates, 10 independent streams x 1,000,000 bits

Test	QRNG (Aer sim)	Mersenne Twister	ChaCha20	os.urandom
Monobit	100%	100%	100%	100%
Block Freq (M=128)	100%	100%	100%	100%
Runs	100%	90%	100%	90%

Test	QRNG (Aer sim)	Mersenne Twister	ChaCha20	os.urandom
DFT Spectral	100%	100%	100%	100%
Byte Chi-Square	100%	100%	100%	100%
Mean Shannon (bits/bit)	0.99981	0.99982	0.99983	0.99981
Mean key bias (ideal ~0.0249)	0.02484	0.02485	0.02499	0.02491

Expected pass rate for a perfect source is 99%. With only 10 streams, a single failure shows as 90% (Runs for MT19937 and os.urandom); this is consistent with chance. The reference sources (ChaCha20, os.urandom) show the same 1-2% failures as the simulated QRNG, so the tests are calibrated and the QRNG sits in the same range.

5. Security-relevant findings

Generation time for 10,000 bits (median of 7 runs, this machine): QRNG simulation 0.023 s; MT19937 17 microseconds; ChaCha20 8 microseconds; os.urandom 5 microseconds. The simulator is about 1000x slower. Absolute values depend on hardware.
MT19937 state recovery: from 624 consecutive 32-bit outputs the generator state is reconstructed and the next 1,000 outputs are predicted correctly (yes). Passing statistical tests says nothing about unpredictability; MT19937 must not generate keys.
Bias removal: on a deliberately biased 52%-ones source, the von Neumann extractor moves the ones-ratio from 0.5199 to 0.4995 (monobit p from ~0 to 0.29) but keeps only 25% of bits. SHA-256 conditioning is also implemented; it hides bias but adds no entropy.
Key quality: uniqueness is 1.00 for all sources, mean bias matches the ideal ~0.0249, so no source shows a key-generation defect at these sizes.

6. Limitations and next steps

• **Real hardware:** run `run_ibm.py` on an IBM backend. Raw hardware bits are expected to lean toward 0 (readout error, T1 decay); compare raw vs von Neumann vs SHA-256-conditioned. This section will be added with the measured results.
• Only 5 of the 15 NIST SP 800-22 tests are implemented; run the full suite, Dieharder or PractRand on 10^8 or more bits.
• Add NIST SP 800-90B min-entropy estimators; the simple per-byte min-entropy here is only a rough guide.
• No claim of quantum advantage or true randomness is made for the simulated QRNG.